

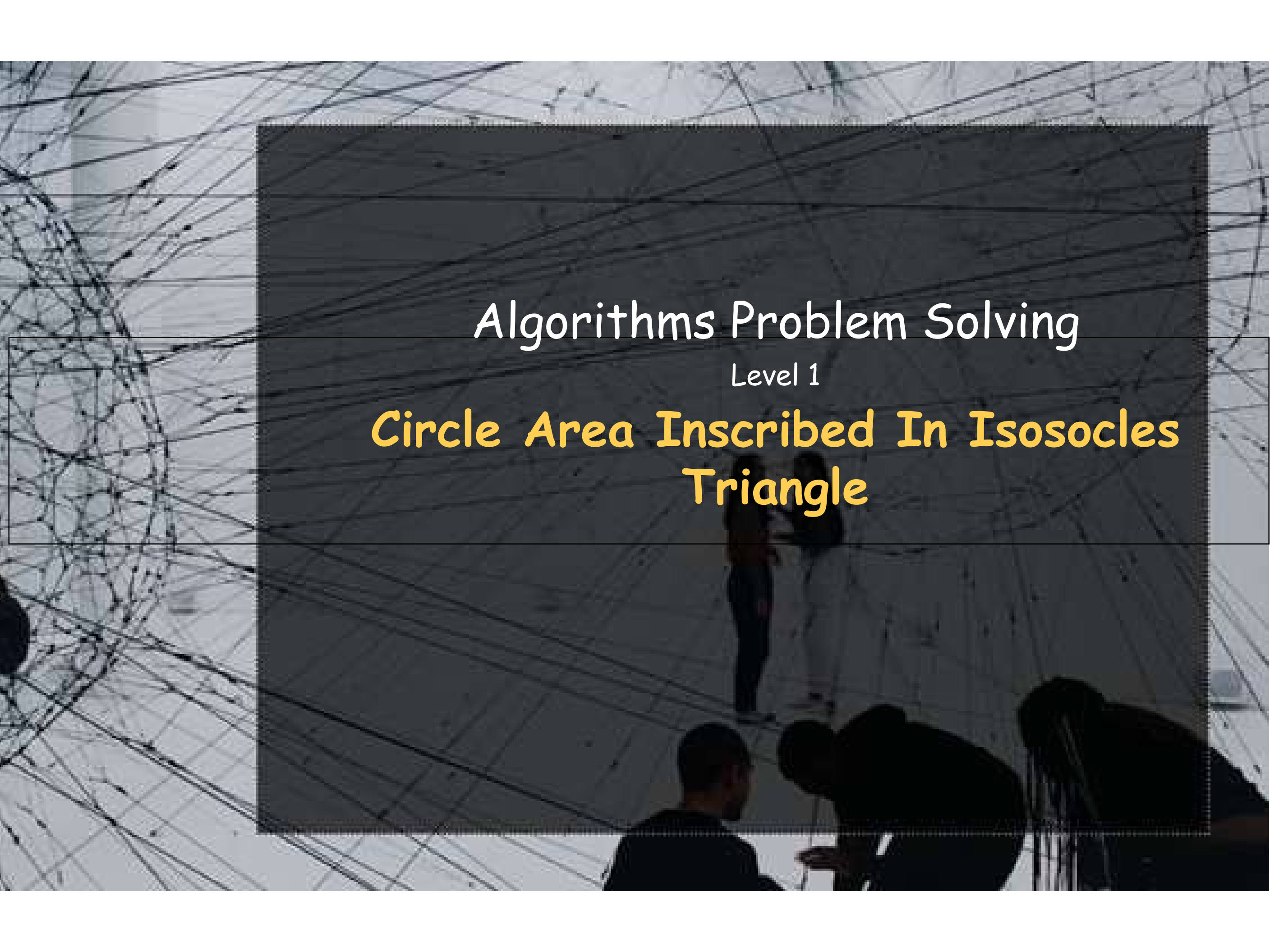
Algorithms Problem Solving Level 1

Circle Area Inscribed In Isosocles Triangle

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Circle Area Inscribed In Isosocles Triangle



🔍 Definition

- ◆ This program calculates the area of a circle inscribed in an isosceles triangle.
- ◆ The triangle has two equal sides A and a base B .
- ◆ The program uses a mathematical formula involving the triangle's equal sides and base, along with π and $\text{pow}()$.
- ◆ The circle is the incircle of the isosceles triangle, meaning it touches all three sides of the triangle.

Explanation

- ◆ The `readTriangleData()` function is intended to read the two equal sides A and the base B .
- ◆ The `CircleAreaByIsoscalesTriangle()` function calculates the circle area using:
- ◆ $\text{Area} = (\pi \times B^2 / 4) \times ((2 \times A - B) / (2 \times A + B))$
- ◆ `pow(B, 2)` calculates B^2 .
- ◆ `M_PI` represents π .
- ◆ The expression $((2 \times A - B) / (2 \times A + B))$ adjusts the result according to the geometry of the isosceles triangle.
- ◆ The `printResult()` function receives the calculated area and displays it.
- ◆ In `main()`, the program reads the triangle data and passes the values to `CircleAreaByIsoscalesTriangle()`.

Steps in main()

1. Read the length of the two equal sides A .
2. Read the length of the base B .
3. Calculate B^2 .
4. Multiply B^2 by π and divide by 4.
5. Calculate the fraction using A and B .
6. Multiply the two parts together.
7. Return the circle area.
8. Print the calculated area.

💡 Example

◆ If the user enters:

Equal side A: 20

Base B: 10

◆ The program calculates:

$$\text{Area} = (\pi \times 10^2 / 4) \times ((2 \times 20 - 10) / (2 \times 20 + 10))$$

◆ The result is approximately 47.1239.

➡ Result

Circle Area = 47.1239

Note

- ◆ For an isosceles triangle, the two equal sides must be long enough to form a valid triangle with the base.
- ◆ The formula calculates the area of the inscribed circle directly without separately calculating the triangle's height, perimeter, or inradius.

⊘ Warning

- ◆ There is a variable-shadowing problem in `readTriangleData()`.
- ◆ The function parameters are `float& A` and `float& B`, but the function declares new local variables:
 - ◆ `int A, B;`
 - ◆ These local variables hide the reference parameters, so the values read from the user are stored in the local int variables instead of the variables in `main()`.
 - ◆ Because of this, the current program does not correctly pass the entered values back to `main()`.

- ◆ The function should be written as:

```
void readTriangleData(float& A, float& B)
{
    cout << "Please enter the length of the equal sides A" << endl;
    cin >> A;
    cout << "Please enter the length of the base" << endl;
    cin >> B;
}
```

- ◆ Also, `CircleAreaByIsoscalesTriangle()` uses int parameters while `main()` uses float variables. Using float parameters would preserve decimal input.



Comparison

▨ Equal Sides A

- ✓ The two sides of the isosceles triangle have the same length.
- ✓ A is used in both parts of the formula.
- ✓ A affects the size of the inscribed circle.

▤ Base B

- ✓ B is the third side of the triangle.
- ✓ B^2 is part of the formula.
- ✓ B also affects the ratio used in the calculation.

✦✦ Summary

- ✓ An isosceles triangle has two equal sides and one base.
- ✓ The program calculates the area of its inscribed circle directly.
- ✓ `pow(B, 2)` calculates the square of the base.
- ✓ The formula uses both the equal side *A* and base *B*.
- ✓ Reference parameters must not be hidden by local variables with the same names.
- ✓ Matching float parameter types helps preserve decimal input values.

Thank you 😊